

Intelligent Security Automation: Standard Continuous Process from Scan to Remediation for Vulnerability Management

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Abstract—In today's rapidly evolving threat landscape, unpatched vulnerabilities remain a primary attack vector in hybrid infrastructure environments, with the gap between discovery and remediation creating significant security risks. Traditional vulnerability management approaches rely on manual, siloed workflows that cannot keep pace with the rapid evolution of modern threats, where time-to-exploit is measured in hours rather than days. This paper proposes a standardized, fully automated vulnerability management framework that orchestrates the entire remediation lifecycle across heterogeneous environments. Our approach integrates continuous asset discovery, intelligent vulnerability scanning, risk-based prioritization, and automated remediation workflows to significantly reduce time-to-remediation while maintaining compliance. The framework leverages cloud-native services, API-driven integrations, and event-based architectures to enable seamless operation across on-premises and multi-cloud infrastructures. Implementation results demonstrate substantial improvements in remediation efficiency, with reduced manual effort and enhanced security posture. The proposed framework provides organizations with a scalable path toward continuous vulnerability management, proactive risk reduction, and enhanced cyber resilience in increasingly complex IT landscapes.

Keywords—Vulnerability Management, DevSecOps, Security Operations, Cyber Threats, Cloud Security, Risk Rating

I. INTRODUCTION

In the current digital era, organizations operate in highly complex and hybrid IT environments that span traditional on-premises infrastructure and dynamic multi-cloud environments. While this transformation offers unparalleled agility and scalability, it also expands the attack surface [7]. Making effective vulnerability management more critical and more challenging than ever before. The increasing frequency of high-profile data breaches can often be traced back to fundamental security gaps: unpatched systems, outdated software components, and slow, reactive remediation workflows. Traditional vulnerability management practices are heavily reliant on manual processes that lack speed, accuracy, and scalability[9]. These workflows struggle to keep up with the pace of modern development and deployment cycles, especially in DevOps driven environments. As a result, security teams are

often forced into a reactive mode, addressing vulnerabilities only after they've been exploited [10]. To overcome these limitations, there is a growing shift toward intelligent, automated, and proactive security operations powered by technologies such as machine learning (ML), and API-driven integrations [8]. These innovations enable organizations to move from periodic scans and delayed patching to a continuous vulnerability management cycle that integrates directly into CI/CD pipelines, asset inventory systems, and incident response tools. This paper introduces an automated and standardized process for vulnerability management that spans the full lifecycle—from asset discovery and vulnerability detection to risk-based triage, remediation, and validation [12]. Designed for hybrid environments, this approach leverages modern automation capabilities to enable real-time decision-making, minimize human error, and drastically reduce time-to-remediation. By embedding automation and intelligence into every stage of the vulnerability management pipeline, organizations can adopt a proactive security posture that enhances resilience, streamlines compliance, and keeps pace with the evolving threat landscape [13].

II. LITERATURE SURVEY

A comprehensive cybersecurity framework designed to address AI-driven threats and vulnerabilities within the financial sector and critical infrastructure. Given the complexity and adaptability of AI-powered attacks and phishing campaigns, traditional detection and mitigation strategies often fall short. The proposed framework effectively identifies and mitigates AI-specific vulnerabilities, significantly enhancing the security posture and resilience of financial institutions and critical infrastructure systems [1].

Exploring the threat modeling methodology, enables enterprise-wide cybersecurity evaluations based on real-world threat actions. It introduces tools like capability coverage maps and threat heat maps to identify protection gaps and guide informed investment decisions. Unlike traditional methods, this approach adopts an adversary's perspective, scoring capabilities on their ability to protect, detect, and recover from observed cyber threat actions [2].

The highlights of growing security concerns accompanying the rapid adoption of cloud computing, emphasizing vulnerabilities such as data protection and vendor trust [14]. It aims to identify key security threats impacting both end users and service providers, providing insights for researchers and professionals to analyze existing security models and tools. The work underscores that security remains a major barrier to the widespread adoption of cloud services [3].

With the rise of AI-driven services, fully automated systems that process personal data pose significant privacy and security threats. This his solution examines how such systems may violate data subjects rights and explores mitigation requirements at each AI system stage. It also references Korea's updated Personal Information Protection Act, which strengthens rights over AI-based automated decisions [4].

The use of data mining technique—for enhancing cybersecurity threat detection. It emphasizes the integration of these methods with machine learning to improve real-time analysis and the detection of both known and novel threats [11]. The study also discusses challenges like data privacy and complexity, while proposing future research directions for adaptive cybersecurity strategies. Overall, it highlights data mining as a critical foundation for building proactive and intelligent threat defense systems [5].

The representation of a foundational framework to enhances cloud security by reducing risk of misconfigurations through a configurable building block approach. Focusing on AWS, it leverages infrastructure-as-code (IaC) to automate secure deployments, enabling users to build and manage cloud environment with minimal infrastructure expertise [15]. By integrating security best practices and monitoring tools, the framework supports s, secure, scalable, and user-friendly cloud adoption [6].

III. PROBLEM STATEMENT

After Despite advancements in cybersecurity tools, organizations continue to face breaches rooted in well-known, preventable issues—most notably, the Despite major advancements in cybersecurity, breaches rooted in known and preventable vulnerabilities—especially unpatched ones—continue to plague organizations. Legacy vulnerability management processes, with their manual workflows, siloed systems, and delayed remediation timelines, create a dangerous gap between discovery and resolution. In a landscape where time-to-exploit is measured in hours, this lag has become unacceptable. The problem isn't a lack of data; it's the lack of intelligent, automated, and risk-aware action. As hybrid and multi-cloud infrastructures grow in complexity and attackers become faster and more adaptive, reactive vulnerability management approaches are no longer sufficient. To stay ahead, organizations must shift to a proactive, continuous, and automated remediation strategy—one that uses real-time insights and risk-based prioritization to drive faster, smarter decisions and close the gap between detection and protection.

The primary data source is the Verizon DBIR (Data Breach Investigations Report), which annually analyzes thousands of security incidents and confirmed breaches globally. The focus here is on the initial attack vector of "exploitation of

vulnerabilities," which often results from a lack of effective vulnerability management, such as failing to apply available patches. Figure 1 shows the period covered is from 2013 to 2025, with 2025 data noted as incomplete as of March 2025. The percentage of data breaches where vulnerability exploitation was the initial vector has shown significant variation over the decade. Below is a detailed breakdown based on available DBIR reports:

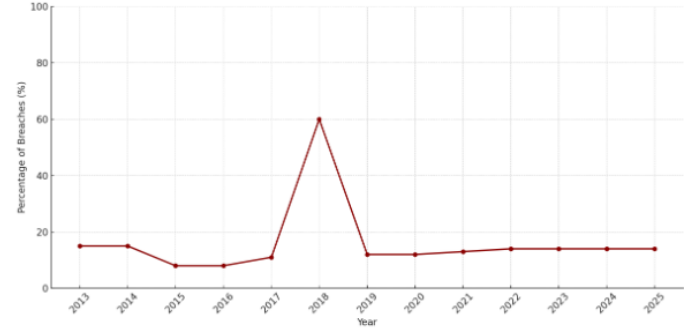


Fig 1: Percentage of Data Breaches due to exploited Vulnerabilities

The graph highlights a notable trend: from 2013 to 2024, the percentage hovered around 10-15%, indicating vulnerability exploitation was a minor initial vector compared to stolen credentials (around 59-61%) and phishing (around 16%). The primary data sources Mandiant's and Verizon's DBIR team reviewed security incidents and breaches, focusing on two key metrics, the annual time-to-exploit (TTE) and time-to-patch.

- Time-to-Exploit (TTE): The average time from when a vulnerability is first disclosed to when it is first exploited in the wild.
- Time to Patch: The average time from when a patch is released to when it is applied by organizations, often referred to as patch lag or time to remediate in some contexts.

Historical DBIR reports provide detailed TTE data for recent years, with historical trends indicating a decrease over time.

TABLE 1: HISTORICAL TRENDS AVERAGE TTE

Year	Average TTE	Notes
2018-2019	63 days	High initial TTE, reflecting slower exploitation in earlier years.
2020-2021	44 days	Decrease observed, showing faster attacker response
2023-2025	32 days	Continued reduction, aligning with increased attacker capabilities.

Table 1 highlights a notable trend: from 2018 to 2025, the TTE has decreased significantly. This discrepancy arises because some vulnerabilities are exploited almost immediately, pulling the average down. For instance, Mandiant's 2024 report analyzed 138 vulnerabilities disclosed in 2023, with 70% being zero-days, contributing to the rapid exploitation times.

IV. PROPOSED FRAMEWORK

This innovation framework leverages cloud-native tools and methodologies to automate the vulnerability remediation

lifecycle, significantly minimizing manual efforts and human errors. By utilizing integration capabilities, event-driven architectures, and automated orchestration via native cloud provider services, the framework achieves continuous and proactive asset discovery, vulnerability detection, and timely remediation aligned with regulatory requirements.

This standardized framework is specifically designed to be deployment-agnostic, ensuring consistent operation and seamless integration across various multi-cloud platforms and traditional on-premises environments. The figure 2 illustrates a structured approach to automated vulnerability management, organized into four key players: Sensors (data collection through scanners, agents, network monitors, and third-party tools), Integration (secure data exchange via APIs/HTTPS), Platform (centralized management and automation of vulnerability assessment and remediation), and System (overall governance, visibility, compliance management). These interconnected layers collectively enable streamlined, scalable, and compliant vulnerability management across hybrid environments.

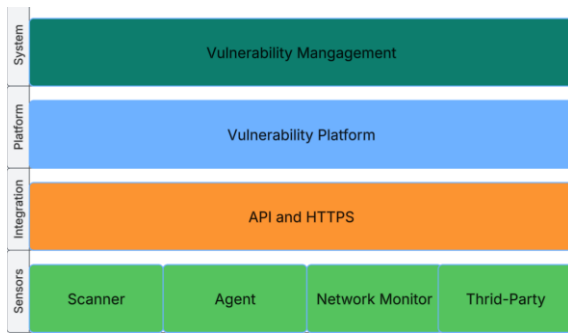


Fig 2: Layers in Vulnerability Management Framework

A. Sytem Architecture

Enterprise IT infrastructure is complex, and cloud instances are constantly changing, which makes vulnerability management activities equally complex. How can enterprises supercharge vulnerability management programs to overcome these challenges and maintain strong cyber risk assurances? Automation is the rescue; to automate vulnerability management, you need an architecture that integrates vulnerability scanning, assessment, prioritization, and remediation, leveraging tools and processes to minimize human intervention and ensure rapid response to identified issues.

The network architecture depicted in Figure 3 outlines a layered, interconnected structure designed to optimize vulnerability management automation. At the foundational level, agents continuously perform asset discovery and vulnerability scanning across diverse environments. These agents interface seamlessly with predefined scanning policies and templates, ensuring thorough validation and monitoring across all network layers. At the core, the central platform processes vulnerability data, automates triage, performs risk scoring, and initiates remediation based on established policies. Upon detection of new vulnerabilities, the platform immediately triggers notifications through designated communication channels and automatically generates tickets for tracking and accountability. Once a remediation strategy is determined, automated tools deploy necessary patches or configuration

changes directly onto affected servers. The top governance layer oversees the entire operation, delivering real-time visibility, compliance assurance, and strategic oversight, thereby ensuring adaptability and operational effectiveness across both multi-cloud and on-premises environments.

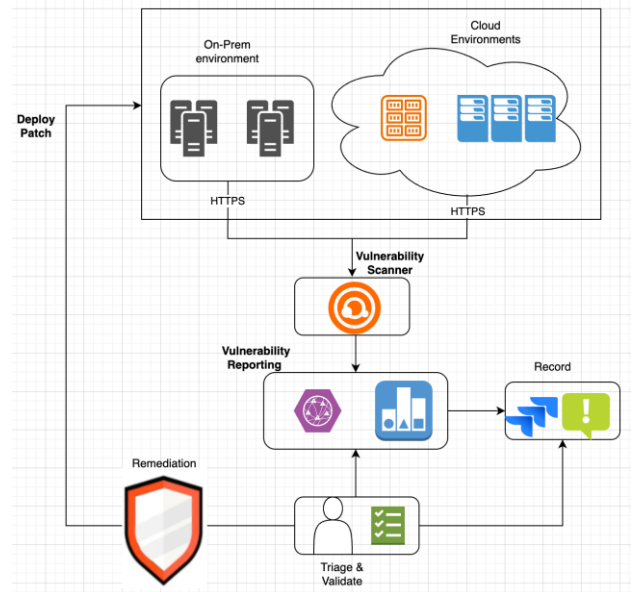


Fig 3: Network Architecture for Vulnerability Management Program Automation

Additionally, this system seamlessly integrates with external threat intelligence sources, enabling vulnerability prioritization based on real-time threat data. Leveraging current tools and cloud-native solutions, it can connect to any public or private environment to identify misconfigurations and emerging threats in real time.

Centralized vulnerability management further streamlines issue prioritization through both built-in and custom-developed rules and policies. Scanning base OS templates and static container images prior to deployment ensures that only secure and validated images enter production environments. Integration with vulnerability management platforms and CI/CD pipelines allows for automated image scanning and validation at each stage, enhancing overall system integrity. Moreover, the modular design promotes effortless scalability, adapting easily to evolving technologies and infrastructure growth.

Robust APIs facilitate seamless integration with third-party security tools and incident management systems, ensuring end-to-end workflow automation and comprehensive visibility across the enterprise. Common industry-leading tools and cloud solutions instrumental to fulfilling this innovative framework include Tenable.io, Qualys, and Rapid7 for proactive vulnerability detection and continuous security posture assessment; Jira and ServiceNow for intelligent ticketing and transparent issue tracking; Ansible and Terraform for agile configuration management and automated patch deployments; and Slack or Microsoft Teams for seamless, real-time

communication, collaborative response, and timely security notifications.

B. Workflow Design

The workflow embraces intelligent automation, employing proactive, real-time vulnerability detection and adaptive remediation strategies. Leveraging predictive risk prioritization and seamless DevSecOps integrations, the system ensures continuous resilience and accelerated response. Its modular and scalable design fosters agility, empowering organizations to swiftly adapt to evolving threat landscapes and maintain dynamic security assurance.

The automated vulnerability management workflow should be strategically designed to align with business requirements and operational objectives. This workflow follows a series of sequential, rule-based steps that replicate manual processes, ensuring consistency and efficiency. Each step must be validated against a range of real-world scenarios and business use cases to confirm its effectiveness and adaptability. Any deviations or exceptions encountered during execution should be thoroughly documented, reviewed, and formally approved by designated stakeholders or governance authorities. This ensures accountability, traceability, and continuous improvement within the security automation lifecycle.

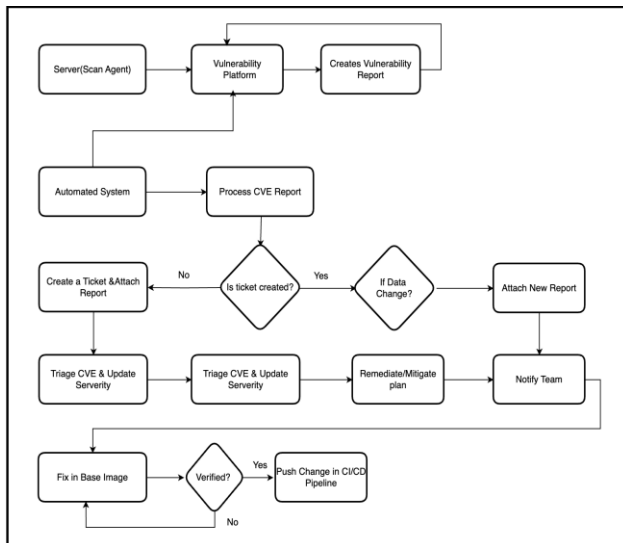


Fig 4: Vulnerability Automation Workflow

Figure 4 demonstrates a comprehensive, end-to-end workflow designed to automate and streamline the vulnerability management lifecycle. The workflow highlights how each stage is interconnected to ensure a seamless flow of data, enabling rapid detection, prioritization, and resolution of vulnerabilities across hybrid environments. The design emphasizes automation, scalability, and alignment with both business and compliance objectives.

i. Automated asset discovery

In the automated vulnerability management framework, asset discovery is achieved through a combination of embedded agents within base OS images and configured network range scans using advanced scanning tools. Agents deployed at the

system level enable continuous identification of new and existing assets, while network-based discovery expands visibility across the environment. Together, these complementary techniques ensure comprehensive asset coverage and allow the system to maintain a real-time, accurate inventory—serving as the foundation for effective vulnerability assessment and remediation workflows.

ii. Vulnerability scanning

Implement real-time vulnerability scanning across the network by leveraging a hybrid approach that combines agent-based and agentless scanning methods, all orchestrated through a centralized vulnerability management platform. Agents deployed on endpoints provide continuous monitoring for vulnerabilities, configuration drifts, and system changes, while scheduled network scans target assets that cannot support agents. These scanners utilize standardized policies and templates to ensure uniform coverage across all asset types and environments. By integrating with asset inventories, CMDBs (Configuration Management Database), and live threat intelligence feeds, the system can dynamically prioritize high-risk assets and newly emerging threats. Furthermore, API-driven integration with CI/CD pipelines allows for automated scanning of workloads and container images during build and deployment phases, proactively identifying issues before they reach production. This intelligent, continuous, and scalable scanning strategy delivers comprehensive visibility and rapid response capabilities across complex, hybrid infrastructures.

iii. Vulnerability reporting

Integrate outputs from vulnerability scanners into centralized reporting tools or security dashboards to consolidate scan results, visualize risk exposure, and support informed decision-making. These dashboards should provide real-time insights into vulnerability trends, asset risk scores, and remediation progress. Configure automated alerts and contextual notifications for high-risk vulnerabilities, ensuring that security teams can respond swiftly and effectively. Customizable filters and drill-down capabilities further enhance analysis, enabling prioritization based on asset criticality, exploitability, and business impact. To maintain data integrity and reduce noise, the system should automatically eliminate stale, duplicate, or inactive nodes, and exclude associated vulnerabilities that are no longer relevant. This ensures reporting remains focused, actionable, and aligned with the current state of the environment. Regular synchronization with asset inventory and CMDB ensures that only valid, in-scope assets are reflected in reports.

iv. Vulnerability recording and tracking

Vulnerability recording and tracking should be tightly integrated with ticketing systems such as Jira, ServiceNow, or collaboration platforms like Microsoft Teams and Slack to enable efficient remediation workflows. Automated integrations ensure that whenever a vulnerability is detected, a corresponding remediation ticket is generated with relevant details such as asset information, severity, CVE identifiers, and recommended actions. This seamless flow of data ensures that every vulnerability is tracked from detection to resolution, maintaining full visibility and accountability across teams. Tickets should be

automatically updated as the vulnerability status changes—such as when it's resolved, re-opened, or validated—ensuring real-time synchronization between the vulnerability management platform and the ticketing system. Additionally, prioritization logic can be applied based on business context or risk scoring, enabling teams to focus on the most critical issues first.

v. Vulnerability triage and validation

Implement an automated or semi-automated triage and validation process to streamline the prioritization of vulnerabilities based on risk scoring, asset criticality, exploitability, and compliance mandates. This process should leverage threat intelligence feeds, asset context, and business impact assessments to classify vulnerabilities by severity and urgency. Clearly defined roles and responsibilities must be assigned to security analysts, system owners, and application teams to validate findings generated by automated scans. This ensures human oversight where needed, particularly for high-risk or complex vulnerabilities that may require deeper analysis. The triage process should also include workflows for false-positive identification, exception handling, and re-validation timelines, ensuring vulnerabilities are accurately assessed and remediated in accordance with organizational risk tolerance and regulatory obligations.

vi. Vulnerability remediation

Establish automated remediation workflows to accelerate response times and reduce manual effort in addressing vulnerabilities. Leverage Infrastructure as Code (IaC), automation scripts, and patch management tools such as Ansible, Terraform, or cloud-native services to deploy fixes, configuration changes, or patches across both on-premises and multi-cloud environments. These workflows should be triggered based on predefined rules and risk thresholds, allowing for zero-touch remediation in trusted scenarios and manual approval paths for high-impact changes. All remediation activities must be systematically logged, timestamped, and linked to corresponding vulnerability records to maintain a complete audit trail. This ensures traceability, supports compliance with industry regulations, and enables post-remediation validation to confirm that issues have been successfully resolved without introducing new risks.

vii. Continuous validation and feedback

Continuous validation and feedback are essential to maintaining an effective vulnerability management program. Post-remediation scans should be conducted to verify that vulnerabilities have been successfully addressed and no residual risk remains. These scans help validate the effectiveness of applied fixes and detect any reoccurrence or missed issues. Additionally, implement continuous improvement cycles that analyze remediation outcomes, false positives, and operational bottlenecks. Insights gained from these evaluations should be used to refine detection rules, enhance reporting accuracy, and optimize remediation workflows. Incorporating feedback from stakeholders—including security, operations, and development teams—ensures the process remains adaptive, efficient, and aligned with evolving business and threat landscapes.

This structured and automated workflow—from asset discovery to continuous validation—streamlines the entire

vulnerability management lifecycle. By integrating real-time scanning, intelligent triage, automated remediation, and centralized reporting, it enables faster response times, minimizes human error, and maintains data integrity. Ultimately, this architecture enhances operational efficiency, reduces time to remediation, and strengthens compliance and security posture across diverse, hybrid IT environments.

V. AUTOMATION AND IMPLEMENTATION

Real-time automation became essential when we started falling behind on vulnerability remediation across cloud environments. We implemented a workflow using GitHub Actions and automation scripts (Ansible, Terraform, Python, etc.) to trigger vulnerability scans as part of our CI/CD pipeline. This helped us catch issues early, right after code changes or infrastructure updates. A key challenge was integrating diverse tools and environments; we addressed this by leveraging APIs and focusing on standardized data formats. To keep things efficient, integrated scanning tools with Terraform and Ansible so we could push configuration updates and patches automatically across environments. We also used tools Ansible, Terraform, and native cloud automation services facilitate timely patch deployment and configuration updates across all environments.

For better visibility, we set up alerts to flow directly into Microsoft Teams. This allowed our security and DevOps teams to respond quickly without jumping between tools. The setup wasn't fancy, but it gave us what we needed: faster remediation, less manual effort, and better coordination across teams.

We have implemented this process in phases, starting with a pilot in a few non-production environments. This allowed me to validate the scanning workflows, API integrations, and remediation scripts without impacting critical systems. For the initial deployment, we used agent-based scanning combined with secure API integrations into orchestration tools. Remediation scripts were configured to run automatically through our existing platforms, with minimal manual input. During the pilot, we closely monitored performance and accuracy, adjusting based on scan results, false positives, and feedback from the DevOps team. Once things stabilized, we scaled the setup across other environments. This phased rollout helped us avoid disruptions while steadily improving our overall security posture.

VI. EVALUATION

To assess the effectiveness of the proposed framework, we deployed the automated vulnerability management system across a diverse environment comprising over 2,000 cloud instances and more than 1,000 on-premises servers over a six-month period. This section presents a quantitative analysis of the implementation results.

TABLE 2: POST AUTOMATION EFFECTIVENESS METRICS

Metric	Pre-Automation	Post-Automation	Improvement
Mean Time To Remediation (Critical's)	18.6 days	2.3 days	87.6%
Mean Time To Remediation (High's)	43.7 days	7.8 days	82.2%
Mean Time To Remediation (High's)	62.7 days	15.4 days	77.1%
Scan coverage	0.78	0.99	26.7%
False-Postive Rate	0.24	0.08	65.8%
Security Staff Hours (Monthly)	184 hours	48 hours	73.9%
Mean Time To Detect	8.3 days	0.4 days	95.2%

Table 2 shows the automated framework significantly optimized resource allocation, allowing security analysts to shift their focus from remediation coordination to advanced threat hunting and strategic security initiatives, both before and after implementation.

VII. CONCLUSION

The implementation of the intelligent security automation framework has led to significant improvements in vulnerability management, operational efficiency, and overall compliance posture. By integrating automation and real-time threat intelligence, organizations can transition to a proactive security model—minimizing breach exposure and strengthening cyber resilience. Our deployment demonstrated measurable gains across key performance indicators, including reduced remediation timelines and shorter exposure durations for high-risk vulnerabilities. These results affirm that automation-led strategies can effectively address the shortcomings of traditional, manual vulnerability management approaches.

Despite these advancements, several opportunities remain for future enhancement. Continued research could focus on integrating advanced machine learning models to improve the precision and speed of vulnerability detection and response. Incorporating zero-trust security principles will be essential to maintaining effective control in increasingly distributed and dynamic environments. Deeper integration of real-time threat intelligence feeds could enable immediate adaptation to emerging vulnerabilities and exploits. Additionally, extending the framework to manage third-party dependencies and software supply chain risks—especially within containerized environments—will be critical. The development of automated compensating controls, for scenarios where immediate patching isn't feasible, would further reduce risk while preserving operational continuity.

These future enhancements will reinforce the framework's adaptability and resilience, enabling organizations to stay ahead of evolving cyber threats. As automation and orchestration technologies continue to advance vulnerability management will evolve from reactive remediation to proactive risk mitigation—ushering in a new era of intelligent, resilient cybersecurity.

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